

Chapter: Life Processes

1. Explain the process of nutrition in Amoeba.
2. Describe the alimentary canal of man.
3. Explain the process of breathing in humans.
4. Explain the importance of soil for plant growth.
5. Draw the diagram of the alimentary canal of man and label the following parts.
Mouth, Oesophagus, Stomach, Intestine
6. How do carbohydrates, proteins and fats get digested in human beings?
7. Explain the mechanism of photosynthesis.
8. Explain the three pathways of breakdown in living organisms.
9. Describe the flow of blood through the heart of human beings.
10. Describe the process of urine formation in the kidneys.

Chapter: How do Organisms Reproduce?

1. Why are budding, fragmentation and regeneration all considered as asexual types of reproduction? With neat diagrams, explain the process of regeneration in Planaria.
2. Write two points of difference between asexual and sexual types of reproduction. Describe why variations are observed in the offspring formed by sexual reproduction.
3. Distinguish between pollination and fertilisation. Mention the site and product of fertilisation in a flower. Draw a neat, labelled diagram of a pistil showing pollen tube growth and its entry into the ovule.
4. Distinguish between a gamete and a zygote. Explain their roles in sexual reproduction.
5. Draw the diagram of a flower and label the four whorls. Write the names of gamete-producing organs in the flower.
6. What is a placenta? Mention its role during pregnancy?
7. What are the various ways to avoid pregnancy? Elaborate on any one method.
8. How does fertilisation take place? Fertilisation occurs once a month. Comment.
9. Reproduction is a phenomenon that is not for the survival of an individual but for the stability of a species. Justify.
10. Describe sexually transmitted diseases and mention the ways to prevent them.

Chapter: Heredity

1. Does the geographical isolation of individuals of a species lead to the formation of a new species? Provide a suitable explanation.
2. Bacteria have a simpler body plan than human beings. Does it mean that human beings are more evolved than bacteria? Provide a suitable explanation.
3. All human races, such as Africans, Asians, Europeans, Americans, and others, may have evolved from a common ancestor. Provide a few pieces of evidence in support of this view.
4. Differentiate between inherited and acquired characters. Give one example for each type.
5. Give reasons why acquired characters are not inherited.
6. Evolution has shown greater stability in molecular structures than in morphological structures. Comment on the statement and justify your opinion.
7. Give the basic features of the mechanism of inheritance.
8. Give reasons for the appearance of new combinations of characters in the F₂ progeny.